

Time: 2 ½ hrs

Total Marks: 80

General Instructions:

1. The question paper consists of 34 questions and is divided into four sections, A, B, C and D
 2. All questions are compulsory.
 3. Section A comprises question numbers 1 to 15. These are multiple choice questions carrying one mark each. You are to select one most appropriate response out of the four provided options.
 4. Section B comprises question numbers 16 to 22. These are SAQs carrying two marks each.
 5. Section C comprises question numbers 23 to 31. These are SAQs carrying four marks each.
 6. Section D comprises question numbers 32 to 34. These are SAQs carrying five marks each.
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SECTION A

- Q.1** The part of the teeth which contains nerves and blood vessels is called (1)
- A. Gum
 - B. Pulp cavity
 - C. Enamel
 - D. Dentine
- Q.2** Earthworm traps oxygen from the soil with the help of (1)
- A. Skin
 - B. Gills
 - C. Lungs
 - D. Spiracles
- Q.3** Monsoon winds which bring rainfall over India are (1)
- A. Summer monsoon winds
 - B. Winter monsoon winds
 - C. Southeast monsoon winds
 - D. Northeast monsoon winds

Q.4 Which horizon of soil is rich in minerals?

- A. A horizon
- B. C horizon
- C. B horizon
- D. R horizon

Q.5 At which stage does the silk moth cut open the cocoon?

(1)

- A. Beginning of pupal stage
- B. Adult stage
- C. Larval stage
- D. End of pupal stage

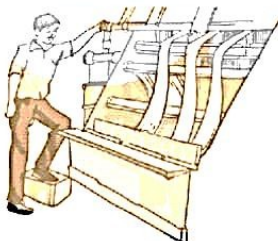
Q.6 Which animal yields wool in South America?

(1)

- A. Angora
- B. Lori
- C. Llama
- D. Camel

Q.7 Which step of wool processing is shown in the following picture?

(1)



- A. Scouring
- B. Shearing
- C. Sorting
- D. Dyeing

Q.8 The taste of lime water (used in whitewashing) is

(1)

- A. Sweet
- B. Sour
- C. Bitter
- D. Salty

Q.9 Which of the following substances change the colour of China rose indicator to green?(1)

- A. Glucose solution
- B. Distilled water
- C. Soda water
- D. Common salt solution

Q.10 The air from the land towards the sea is called

- A. Sea breeze
- B. Land breeze
- C. Land cyclone
- D. Smaller tsunami

Q.11 While reading a thermometer, we should not hold it by the

(1)

- A. Bulb
- B. Glass tube
- C. Kink
- D. Thread

Q.12 The bulb in the thermometer has

(1)

- A. Oxygen
- B. Mercury
- C. Nitrogen
- D. Carbon dioxide

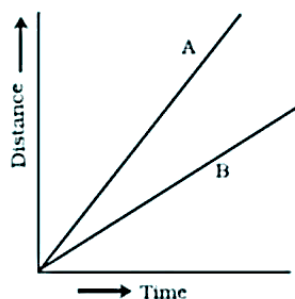
Q.13 One of the most well-known periodic motions is that

(1)

- A. Of sea waves
- B. Of a car
- C. Of a simple pendulum
- D. All of the above

Q.14 Look at the figure given below and choose the correct option from the given options.

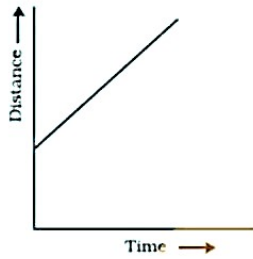
(1)



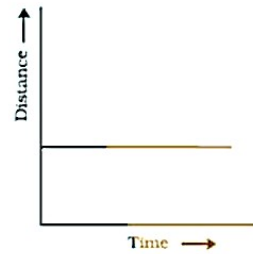
- A. A is moving faster.
- B. B is moving faster.
- C. Both are moving at the same speed.
- D. Speed cannot be calculated.

Q.15 Which one of the following graphs shows a bike with speed 0 m/s?

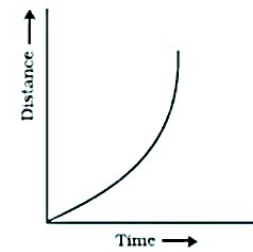
A.



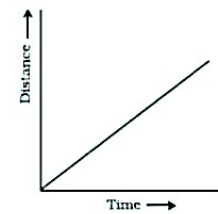
B.



C.



D.



SECTION B



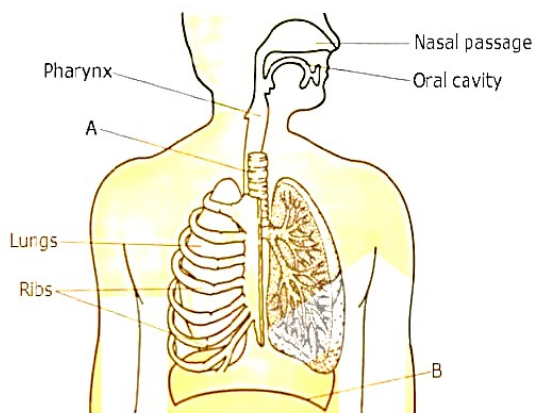
- Q.16** How do leguminous crops replenish the soil? (2)
- Q.17** Why is topsoil best for plant growth? (2)
- Q.18** What are yeasts? How do they respire? (2)
- Q.19** (2)
- (i) When is the shearing of fleece of sheep done?
 - (ii) Why does shearing not hurt the sheep?
- Q.20** Take an artificial (synthetic) silk thread and a pure silk thread. Burn these threads carefully. In a tabular column, write the difference in the smell while burning? (2)
- Q.21** Can a clinical thermometer be used to measure the temperature of boiling milk? (2)
- Q.22** Show the shape of the distance–time graph for motion in the following cases: (2)
- (i) A bike moving with a constant speed.
 - (ii) A car parked on the road side.

SECTION C

- Q.23** (4)
- (i) Draw a neat labelled diagram of Amoeba.
 - (ii) What is the function of pseudopodia in Amoeba?
- Q.24** (4)
- (i) How is the trunk helpful to elephants?
 - (ii) State any two adaptations in penguins which help them to cope with the extreme cold climate in the polar regions.
- Q.25** (4)
- (i) Why are fewer plantations found in deserts?
 - (ii) Why is clay not suitable for growing plants?

Q.26

(i) Identify A and B in the given diagram.



(ii) What is the composition of the air which we inhale and exhale?

Q.27

(4)

(i) What is an indicator?

(ii) Three liquids are given to you. One is sulphuric acid, another is sodium hydroxide and the third is a salt solution. How will you identify them? You have only turmeric paper indicator.

Q.28

(4)

(i) What is a salt?

(ii) What are the different types of salts? Give one example of each.

Q.29

(4)

(i) One end of a wooden spoon is dipped in a cup of ice cream. What change will be observed at its other end?

(ii) Why do we use a handle of wood in a frying pan?

Q.30

(4)

(i) Why are freezers of a refrigerator always at the top?

(ii) Why does hot air rise and cold air sink?

Q.31

(4)

(i) Why do we need to measure time?

(ii) A dog runs behind you for 30 minutes, and the distance covered by the dog is 3 km. What should be your minimum speed if the dog was not able to bite you?

SECTION D



Q.32

(5)

- (i) How do dolphins breathe?
- (ii) Why do humans sneeze?

Q.33 The hair of an animal A is cut from its body in one piece along with a thin layer of its skin in the form of B. This process is called C. The B of this animal is then cleaned by using soap and a lot of water by a process D. The clean B is then converted into yarn E by the process of sorting, dyeing, combing and spinning. (5)

- (i) What could be A and B?
- (ii) Name the processes C and D.
- (iii) What is the yarn E?
- (iv) State one use of E.
- (v) Which breed of animal A is used for producing wool used for making woollen shawls?

Q.34

(5)

- (i) Of a line graph, a pie chart and a bar graph, which one is the most suitable to show
 1. Runs scored in various overs of a cricket match?
 2. Variation of distance covered by a car with time?
 3. Percentage composition of air?
- (ii) State two advantages of drawing distance–time graphs for moving objects.